

Qualification Pack



Precision Machinist - Conventional

QP Code: TNSD/CSC/Q0602

Version: 1.0

NSQF Level: 3.5

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Qualification Pack

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Qualification Pack

TNSD/CSC/Q0602: Precision Machinist - Conventional

Brief Job Description

This qualification comprises various modules essential for the Precision Machinist - Conventional job role. This qualification is designed to enable learners to achieve the following: Operate and maintain conventional machine tools (such as Lathe, Drilling, Milling, and Grinding machines) to manufacture precision parts according to engineering drawings and standard specifications.

Personal Attributes

Apply foundational knowledge in manufacturing technology, engineering measurements, technical mathematics, and applied science relevant to machining operations. Interpret and create engineering drawings using both manual techniques and AutoCAD. Secure employment across a wide range of sectors, including general engineering, manufacturing industries, machine tool industries, and the automobile sector.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [TNSD/CSC/N0610: Perform Basic Manufacturing Operations](#)
2. [TNSD/CSC/N0611: Conduct Engineering Measurements](#)
3. [TNSD/CSC/N0612: Interpret and Create Engineering Drawings](#)
4. [TNSD/CSC/N0613: Create 2D Drawings using AutoCAD](#)
5. [TNSD/CSC/N0614: Apply Technical Mathematics](#)
6. [TNSD/CSC/N0615: Apply Principles of Applied Science](#)
7. [DGT/VSQ/N0104: Employability Skills \(120 Hours\)](#)
8. [TNSD/CSC/N0616: Demonstrate Computer Literacy](#)

Qualification Pack (QP) Parameters

Sector	Capital Goods
Sub-Sector	
Occupation	Precision Machinist-CNC

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Not
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Country	India
NSQF Level	3.5
Credits	40
Aligned to NCO/ISCO/ISIC Code	7233.0102
Minimum Educational Qualification & Experience	10th grade pass
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	15 Years
Last Reviewed On	NA
Next Review Date	07/10/2028
NSQC Approval Date	07/10/2025
Version	1.0
Reference code on NQR	QG-3.5-CG-04454-2025-V1-TNSDC
NQR Version	1.0

Qualification Pack

TNSD/CSC/N0610: Perform Basic Manufacturing Operations

Description

Understand different types of hand tools and machine tools used in tool room/Machine shop. Understand applications of different machine tools used for manufacturing of mechanical parts / assembly. Understand different types of welding processes such as SMAW, GMAW and GTAW

Scope

The scope covers the following :

- Familiarize with the properties & uses of ferrous and non-ferrous metals.
- Knowledge of safety precautions which is to be observed in workshop/ and while working on machines.
- Knowledge of engineering drawing with Limits, fits and GD&T.
- Knowledge of different measuring instruments and gauges used in Machine shop.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Learn different types of ferrous and non-ferrous material used in engineering practices
- PC2.** Understand the Properties and Heat-treatment processes of materials used for engineering purpose.
- PC3.** Understand the safety aspects and PPE used in workshop
- PC4.** Understand the different types of hand tools and bench work like filing, hacksawing, hammering, marking, letter punching, etc.,
- PC5.** Learn the Process of drilling and related operations like tapping, reaming, counter sinking and counter boring.
- PC6.** Understand the Different parts and its functions of conventional tool room machines-Drilling, lathe, Milling and Grinding.
- PC7.** Familiar with Different types of operations like turning,milling,grinding performed on tool room machines.
- PC8.** Understand the different types of work holding devices, Tool holding devices of Drilling, lathe, Milling and grinding.
- PC9.** Understand the Nomenclature of cutting tool angles and re – sharpening
- PC10.** Make right angled block and different types of fits by bench work.
- PC11.** Machine parts as per drawing using different machines and assembling.
- PC12.** Fabricate different parts of a project and assembling

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** Understand the Properties and Heat-treatment processes of materials used for engineering purpose.
- KU2.** Understand the different types of hand tools and bench work like filing, hacksawing, hammering, marking, letter punching, etc.,
- KU3.** Understand the Different parts and its functions of conventional tool room machines-Drilling, lathe, Milling and Grinding.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Familiar with Different types of operations like turning,milling,grinding performed on tool room machines.
- GS2.** Familiar with Different types of operations like turning,milling,grinding performed on tool room machines.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	200	80	90	30
PC1. Learn different types of ferrous and non-ferrous material used in engineering practices	10	-	-	-
PC2. Understand the Properties and Heat-treatment processes of materials used for engineering purpose.	10	-	-	-
PC3. Understand the safety aspects and PPE used in workshop	10	-	-	-
PC4. Understand the different types of hand tools and bench work like filing, hacksawing, hammering, marking, letter punching, etc.,	40	-	-	-
PC5. Learn the Process of drilling and related operations like tapping, reaming, counter sinking and counter boring.	25	-	-	-
PC6. Understand the Different parts and its functions of conventional tool room machines- Drilling, lathe, Milling and Grinding.	25	-	-	-
PC7. Familiar with Different types of operations like turning, milling, grinding performed on tool room machines.	40	-	-	-
PC8. Understand the different types of work holding devices, Tool holding devices of Drilling, lathe, Milling and grinding.	30	-	-	-
PC9. Understand the Nomenclature of cutting tool angles and re - sharpening	10	-	-	-
PC10. Make right angled block and different types of fits by bench work.	-	40	-	10
PC11. Machine parts as per drawing using different machines and assembling.	-	40	-	10
PC12. Fabricate different parts of a project and assembling	-	-	90	10
NOS Total	200	80	90	30

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National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0610
NOS Name	Perform Basic Manufacturing Operations
Sector	Capital Goods
Sub-Sector	
Occupation	Precision Machinist-CNC
NSQF Level	3.5
Credits	26
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

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TNSD/CSC/N0611: Conduct Engineering Measurements

Description

Understand the usage of different measuring instruments Familiarization with different checking gauges used in tool room/manufacturing. Understand the system of limits & fits used in the manufacturing industry.

Scope

The scope covers the following :

- Understand the usage of different measuring tools & familiarization with different checking methods
- Understand the importance of limits and fits in engineering practices.
- Understand the system of GD& T

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Learn about Co ordinate systems and tools for marking
- PC2.** Know difference between checking and measuring and relevant tools used
- PC3.** Understand linear ,angular and bore checking and measuring instruments
- PC4.** Understand limits and fits and reading of tolerance chart
- PC5.** Know different types of fits and its applications
- PC6.** Understand Hole based system and shaft based system
- PC7.** Learn about Interchangeability and standardization

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Know different types of fits and its applications
- KU2.** Understand linear ,angular and bore checking and measuring instruments
- KU3.** Understand limits and fits and reading of tolerance chart

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Learn about Co ordinate systems and tools for marking

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	80	-	-	20
PC1. Learn about Co ordinate systems and tools for marking	10	-	-	-
PC2. Know difference between checking and measuring and relevant tools used	10	-	-	-
PC3. Understand linear ,angular and bore checking and measuring instruments	15	-	-	-
PC4. Understand limits and fits and reading of tolerance chart	15	-	-	-
PC5. Know different types of fits and its applications	10	-	-	-
PC6. Understand Hole based system and shaft based system	10	-	-	20
PC7. Learn about Interchangeability and standardization	10	-	-	-
NOS Total	80	-	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0611
NOS Name	Conduct Engineering Measurements
Sector	Capital Goods
Sub-Sector	
Occupation	Precision Machinist-CNC
NSQF Level	3.5
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0612: Interpret and Create Engineering Drawings

Description

Read and understand engineering drawing so that items can be manufactured as per dimensions and specifications mentioned. Draw Board drawings of parts which are required to be manufactured.

Scope

The scope covers the following :

- Ability to read engineering drawing.
- Understand the different types of threads and its representation.
- Ability to understand simple mechanical assemblies

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Learn the Basics of engineering drawing and free hand sketching
- PC2.** Draw Different geometrical constructions
- PC3.** Draw the types of projection, sectional views with dimensioning
- PC4.** Draw Orthographic views and isometric views
- PC5.** Draw the assembly drawing and part drawing
- PC6.** Draw the Development of different sections
- PC7.** Learn the basics of GD & T
- PC8.** Draw the representation of various types of joints and fasteners
- PC9.** Draw the representation of different threads, weld joints and rivets

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Learn the Basics of engineering drawing and free hand sketching
- KU2.** Learn the basics of GD & T

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Draw the representation of various types of joints and fasteners

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	90	-	10
PC1. Learn the Basics of engineering drawing and free hand sketching	-	5	-	-
PC2. Draw Different geometrical constructions	-	10	-	-
PC3. Draw the types of projection, sectional views with dimensioning	-	15	-	5
PC4. Draw Orthographic views and isometric views	-	15	-	-
PC5. Draw the assembly drawing and part drawing	-	10	-	-
PC6. Draw the Development of different sections	-	10	-	-
PC7. Learn the basics of GD & T	-	5	-	5
PC8. Draw the representation of various types of joints and fasteners	-	10	-	-
PC9. Draw the representation of different threads, weld joints and rivets	-	10	-	-
NOS Total	-	90	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0612
NOS Name	Interpret and Create Engineering Drawings
Sector	Capital Goods
Sub-Sector	
Occupation	Precision Machinist-CNC
NSQF Level	3.5
Credits	2
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0613: Create 2D Drawings using AutoCAD

Description

Understand the method of drawing mechanical parts by using Auto CAD software.

Scope

The scope covers the following :

- Understand the tools and commands used for drawing in AutoCAD.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Introduction to AUTOCAD

PC2. Create a drawing with different command options

PC3. Set up a working area. Understand usage of Grid and Snap mode, zoom functions

PC4. Create and inserting symbols, assigning layers, control line weights

PC5. Create Orthographic views using software

PC6. Draw standards to make the mechanical assembly drawing with details

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Set up a working area Understand usage of Grid and Snap mode, zoom functions

KU2. Create and inserting symbols, assigning layers, control line weights

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Draw standards to make the mechanical assembly drawing with details

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	80	-	20
PC1. Introduction to AUTOCAD	-	5	-	-
PC2. Create a drawing with different command options	-	10	-	-
PC3. Set up a working area. Understand usage of Grid and Snap mode, zoom functions	-	10	-	-
PC4. Create and inserting symbols, assigning layers, control line weights	-	15	-	10
PC5. Create Orthographic views using software	-	15	-	-
PC6. Draw standards to make the mechanical assembly drawing with details	-	25	-	10
NOS Total	-	80	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0613
NOS Name	Create 2D Drawings using AutoCAD
Sector	Capital Goods
Sub-Sector	
Occupation	Precision Machinist-CNC
NSQF Level	3.5
Credits	2
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0614: Apply Technical Mathematics

Description

Understand the concept of basic mathematics used for doing calculations related to engineering application and general purpose

Scope

The scope covers the following :

- Familiar with System of units, conversions, Simple Arithmetic operations of fractions and decimals.
- Understanding the basic concepts of Ratio, proportion and mensuration.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Learn Fundamental and derived units, System of units, conversions, Arithmetic operations of fractions and decimal fractions
- PC2.** Review of Basic concepts of Arithmetic operations under Whole numbers, Common fraction and Decimal Fraction.
- PC3.** Understand the Basic concepts of Square root ,perfect square numbers, Prime factorization method, Application of Pythagora's theorem
- PC4.** Learn to Calculate Ratio, proportion and Percentage related problems
- PC5.** Learn to Find the Area, Perimeter of circles, triangles, Quadrilaterals & Polygons and related problems.
- PC6.** Understand to Evaluate the Volume, lateral surface area, total surface area of 3D shapes.
- PC7.** Learn the Concepts of trigonometry ratios and identities.
- PC8.** Learn to Find the unknown sides of triangle by using the sine, cos, tan ratios, Sine bar , taper ratio related problems.
- PC9.** Understand to find the Solution of triangles using Sine rule, Cosine rule and projection formula and related problems.
- PC10.** Understand the Concepts of Algebra , algebraic expressions , algebraic equations and Factorization by using Algebraic Identities.
- PC11.** Learn to Solve the homogeneous equations by algebraic concepts.
- PC12.** Learn to Perform Addition, subtraction, multiplication and division of matrices.
- PC13.** Learn to find the determinant of matrices, inverse of Matrices, Form the Euclidean form and find the rank of matrices.
- PC14.** Learn to Solve the linear equations by Cramer rule and classification of solution of homogeneous equations

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** Understand the Basic concepts of Square root ,perfect square numbers, Prime factorization method, Application of Pythagoras theorem
- KU2.** Understand to Evaluate the Volume, lateral surface area, total surface area of 3D shapes.
- KU3.** Understand to find the Solution of triangles using Sine rule, Cosine rule and projection formula and related problems.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Learn to find the determinant of matrices, inverse of Matrices, Form the Euclidean form and find the rank of matrices.
- GS2.** Understand to find the Solution of triangles using Sine rule, Cosine rule and projection formula and related problems.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	100	-	-	-
PC1. Learn Fundamental and derived units, System of units, conversions, Arithmetic operations of fractions and decimal fractions	15	-	-	-
PC2. Review of Basic concepts of Arithmetic operations under Whole numbers, Common fraction and Decimal Fraction.	10	-	-	-
PC3. Understand the Basic concepts of Square root ,perfect square numbers, Prime factorization method, Application of Pythagora's theorem	10	-	-	-
PC4. Learn to Calculate Ratio, proportion and Percentage related problems	10	-	-	-
PC5. Learn to Find the Area, Perimeter of circles, triangles, Quadrilaterals & Polygons and related problems.	5	-	-	-
PC6. Understand to Evaluate the Volume, lateral surface area, total surface area of 3D shapes.	5	-	-	-
PC7. Learn the Concepts of trigonometry ratios and identities.	5	-	-	-
PC8. Learn to Find the unknown sides of triangle by using the sine, cos, tan ratios, Sine bar , taper ratio related problems.	10	-	-	-
PC9. Understand to find the Solution of triangles using Sine rule, Cosine rule and projection formula and related problems.	5	-	-	-
PC10. Understand the Concepts of Algebra , algebraic expressions , algebraic equations and Factorization by using Algebraic Identities.	5	-	-	-
PC11. Learn to Solve the homogeneous equations by algebraic concepts.	5	-	-	-
PC12. Learn to Perform Addition, subtraction, multiplication and division of matrices.	5	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Learn to find the determinant of matrices, inverse of Matrices, Form the Euclidean form and find the rank of matrices.	5	-	-	-
PC14. Learn to Solve the linear equations by Cramer rule and classification of solution of homogeneous equations	5	-	-	-
NOS Total	100	-	-	-

Logo
Not
Available



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0614
NOS Name	Apply Technical Mathematics
Sector	Capital Goods
Sub-Sector	
Occupation	Precision Machinist-CNC
NSQF Level	3.5
Credits	2
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0615: Apply Principles of Applied Science

Description

Understand the concepts of basic science used in engineering field.

Scope

The scope covers the following :

- Understand the concepts of work, power, energy, speed, velocity, heat and pressure.
- Understand the concept of magnetism and electricity.
- Understand the type of forces and equilibrium resolution of forces

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Learn the Concepts of units and Quantity , Conversion of units
- PC2.** Understand the concept of work , power & energy and Related problems
- PC3.** Understand Derived quantity and related problems
- PC4.** Understand the Density of different materials and weight calculations
- PC5.** Learn the Concept of Heat and temperature and its effects and related problems
- PC6.** Learn the System of forces and equilibrium and related problems
- PC7.** Learn the principles of Electricity and Magnetism and related calculations
- PC8.** Understand Friction and relation between Stress and strain and related calculations

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand Friction and relation between Stress and strain and related calculations
- KU2.** Understand the concept of work , power & energy and Related problems

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Learn the Concepts of units and Quantity , Conversion of units
- GS2.** Learn the System of forces and equilibrium and related problems

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	100	-	-	-
PC1. Learn the Concepts of units and Quantity , Conversion of units	15	-	-	-
PC2. Understand the concept of work , power & energy and Related problems	15	-	-	-
PC3. Understand Derived quantity and related problems	10	-	-	-
PC4. Understand the Density of different materials and weight calculations	10	-	-	-
PC5. Learn the Concept of Heat and temperature and its effects and related problems	10	-	-	-
PC6. Learn the System of forces and equilibrium and related problems	15	-	-	-
PC7. Learn the principles of Electricity and Magnetism and related calculations	10	-	-	-
PC8. Understand Friction and relation between Stress and strain and related calculations	15	-	-	-
NOS Total	100	-	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0615
NOS Name	Apply Principles of Applied Science
Sector	Capital Goods
Sub-Sector	
Occupation	Precision Machinist-CNC
NSQF Level	3.5
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

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DGT/VSQ/N0104: Employability Skills (120 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** understand the significance of employability skills in meeting the current job market requirement and future of work
- PC2.** identify and explore learning and employability relevant portals
- PC3.** research about the different industries, job market trends, latest skills required and the available opportunities

Constitutional values - Citizenship

To be competent, the user/individual on the job must be able to:

- PC4.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. for personal growth and the nation's progress
- PC5.** follow personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC6.** follow and promote environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC7.** recognize the significance of 21st Century Skills for employment

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- PC8.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life
- PC9.** adopt a continuous learning mindset for personal and professional development

Basic English Skills

To be competent, the user/individual on the job must be able to:

- PC10.** use English as a medium of formal and informal communication while dealing with topics of everyday conversation in different contexts
- PC11.** speak over the phone in English, in an audible manner, using appropriate greetings, opening, and closing statements both on personal and work front
- PC12.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC13.** write short messages, notes, letters, e-mails etc., using accurate English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC14.** identify career goals based on the skills, interests, knowledge, and personal attributes
- PC15.** prepare a career development plan with short- and long-term goals

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC16.** follow verbal and non-verbal communication etiquette while communicating in professional and public settings
- PC17.** use active listening techniques for effective communication
- PC18.** communicate in writing using appropriate style and format based on formal or informal requirements
- PC19.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC20.** • ensure personal behaviour, conduct, and use appropriate communication by taking gender into consideration
- PC21.** empathize with a PwD and aid a PwD, if asked
- PC22.** escalate any issues related to sexual harassment at the workplace in accordance with the POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC23.** identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.
- PC24.** carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook
- PC25.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC26.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

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To be competent, the user/individual on the job must be able to:

- PC27.** operate digital devices and use their features and applications securely and safely
- PC28.** carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.
- PC29.** display responsible online behaviour while using various social media platforms
- PC30.** create a personal email account, send and process received messages as per requirement
- PC31.** carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications
- PC32.** utilize virtual collaboration tools to work effectively

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC33.** identify different types of Entrepreneurship and Enterprises
- PC34.** use research and networking skills to identify and assess opportunities for potential business
- PC35.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC36.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC37.** identify different types of customers
- PC38.** identify and respond to customer requests and needs in a professional manner
- PC39.** use appropriate tools to collect customer feedback
- PC40.** follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

- PC41.** create a professional Curriculum vitae (Résumé)
- PC42.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively
- PC43.** apply to identified job openings using offline /online methods as per requirement
- PC44.** answer questions politely, with clarity and confidence, during recruitment and selection
- PC45.** identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** need for employability skills
- KU2.** different learning and employability related portals
- KU3.** various constitutional and personal values
- KU4.** different environmentally sustainable practices and their importance
- KU5.** Twenty first (21st) century skills and their importance

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- KU6.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up
- KU7.** importance of career development and setting long- and short-term goals
- KU8.** Do's and don'ts of effective communication
- KU9.** POSH Act
- KU10.** inclusivity and its importance
- KU11.** different types of disabilities and appropriate verbal and non-verbal communication and behaviour towards PwD
- KU12.** different types of financial institutes, products, and services
- KU13.** components of salary and how to compute income and expenditure
- KU14.** importance of maintaining safety and security in offline and online financial transactions
- KU15.** different legal rights and laws
- KU16.** different types of digital devices and the procedure to operate them safely and securely
- KU17.** how to create and operate an e- mail account
- KU18.** use applications such as word processors, spreadsheets etc.
- KU19.** different types of Enterprises and ways to identify business opportunities
- KU20.** types and needs of customers
- KU21.** how to apply for a job and prepare for an interview
- KU22.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and write different types of documents/instructions/correspondence in English and other languages
- GS2.** communicate effectively using appropriate language in formal and informal settings
- GS3.** behave politely and appropriately with all to maintain effective work relationship
- GS4.** how to work in a virtual mode, using various technological platforms
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the current job market requirement and future of work	-	-	-	-
PC2. identify and explore learning and employability relevant portals	-	-	-	-
PC3. research about the different industries, job market trends, latest skills required and the available opportunities	-	-	-	-
<i>Constitutional values – Citizenship</i>	2	1	-	-
PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. for personal growth and the nation's progress	-	-	-	-
PC5. follow personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC6. follow and promote environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	3	-	-
PC7. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC8. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
PC9. adopt a continuous learning mindset for personal and professional development	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. use English as a medium of formal and informal communication while dealing with topics of everyday conversation in different contexts	-	-	-	-
PC11. speak over the phone in English, in an audible manner, using appropriate greetings, opening, and closing statements both on personal and work front	-	-	-	-
PC12. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC13. write short messages, notes, letters, e-mails etc., using accurate English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-
PC14. identify career goals based on the skills, interests, knowledge, and personal attributes	-	-	-	-
PC15. prepare a career development plan with short- and long-term goals	-	-	-	-
<i>Communication Skills</i>	2	3	-	-
PC16. follow verbal and non-verbal communication etiquette while communicating in professional and public settings	-	-	-	-
PC17. use active listening techniques for effective communication	-	-	-	-
PC18. communicate in writing using appropriate style and format based on formal or informal requirements	-	-	-	-
PC19. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC20. • ensure personal behaviour, conduct, and use appropriate communication by taking gender into • consideration	-	-	-	-
PC21. empathize with a PwD and aid a PwD, if asked	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. escalate any issues related to sexual harassment at the workplace in accordance with the POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC23. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.	-	-	-	-
PC24. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook	-	-	-	-
PC25. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC26. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	2	3	-	-
PC27. operate digital devices and use their features and applications securely and safely	-	-	-	-
PC28. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.	-	-	-	-
PC29. display responsible online behaviour while using various social media platforms	-	-	-	-
PC30. create a personal email account, send and process received messages as per requirement	-	-	-	-
PC31. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications	-	-	-	-
PC32. utilize virtual collaboration tools to work effectively	-	-	-	-
<i>Entrepreneurship</i>	2	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC33. identify different types of Entrepreneurship and Enterprises	-	-	-	-
PC34. use research and networking skills to identify and assess opportunities for potential business	-	-	-	-
PC35. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC36. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC37. identify different types of customers	-	-	-	-
PC38. identify and respond to customer requests and needs in a professional manner	-	-	-	-
PC39. use appropriate tools to collect customer feedback	-	-	-	-
PC40. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	4	-	-
PC41. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC42. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC43. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC44. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC45. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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Not
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Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0104
NOS Name	Employability Skills (120 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	6
Credits	4
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0616: Demonstrate Computer Literacy

Description

Understand the usage of computer and its application for personal and professional use

Scope

The scope covers the following :

- Knowledge to work on a computer by using MS office
- Knowledge of internet applications with E-mail and mobile.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the computers and its applications
- PC2.** Learn the operating systems and basic operations
- PC3.** Practice working with MS – Word
- PC4.** Practice working with MS – Excel
- PC5.** Practice working with MS - Power Point Presentation
- PC6.** Do WEB Browsing using internet
- PC7.** Practice communicating with Email and mobile applications

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the computers and its applications
- KU2.** Learn the operating systems and basic operations

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Learn the operating systems and basic operations
- GS2.** Understand the computers and its applications

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	80	-	20
PC1. Understand the computers and its applications	-	5	-	5
PC2. Learn the operating systems and basic operations	-	10	-	-
PC3. Practice working with MS – Word	-	10	-	-
PC4. Practice working with MS – Excel	-	10	-	5
PC5. Practice working with MS - Power Point Presentation	-	20	-	5
PC6. Do WEB Browsing using internet	-	10	-	-
PC7. Practice communicating with Email and mobile applications	-	15	-	5
NOS Total	-	80	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0616
NOS Name	Demonstrate Computer Literacy
Sector	Capital Goods
Sub-Sector	
Occupation	Precision Machinist-CNC
NSQF Level	3.5
Credits	2
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Assessment System Overview:

- Batches assigned to the assessment agencies for conducting the assessment on SIP or email
- Assessment agencies send the assessment confirmation to VTP/TC looping TNSDC
- Assessment agency deploys the ToA certified Assessor for executing the assessment
- TNSDC monitors the assessment process & records

2. Testing Environment:

- Check the Assessment location, date and time
- If the batch size is more than 30, then there should be 2 Assessors.
- Check that the allotted time to the candidates to complete Theory & Practical Assessment is correct.

3. Assessment Quality Assurance levels/Framework:

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- Question bank is created by the Subject Matter Experts (SME) are verified by the other SME
- Questions are mapped to the specified assessment criteria
- Assessor must be ToA certified & trainer must be ToT Certified

4. Types of evidence or evidence-gathering protocol:

- Time-stamped & geotagged reporting of the assessor from assessment location
- Centre photographs with signboards and scheme specific branding

5. Method of verification or validation:

- Surprise visit to the assessment location

6. Method for assessment documentation, archiving, and access

- Hard copies of the documents are stored

On the Job:

1. Each module will be assessed separately.

2. The candidate must score 60% in each module to successfully complete the OJT.

3. Tools of Assessment that will be used for assessing whether the candidate is having desired skills and etiquette of dealing with customers, understanding needs & requirements, assessing the customer and perform Soft Skills effectively:

- Videos of Trainees during OJT

4. Assessment of each Module will ensure that the candidate is able to:

- Effective engagement with the customers
- Understand the working of various tools and equipment

Minimum Aggregate Passing % at QP Level : 60

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
TNSD/CSC/N0610.Perform Basic Manufacturing Operations	200	80	90	30	400	40
TNSD/CSC/N0611.Conduct Engineering Measurements	80	0	0	20	100	10
TNSD/CSC/N0612.Interpret and Create Engineering Drawings	0	90	0	10	100	10
TNSD/CSC/N0613.Create 2D Drawings using AutoCAD	0	80	0	20	100	10
TNSD/CSC/N0614.Apply Technical Mathematics	100	0	0	0	100	10
TNSD/CSC/N0615.Apply Principles of Applied Science	100	0	0	0	100	10
DGT/VSQ/N0104.Employability Skills (120 Hours)	20	30	-	-	50	5
TNSD/CSC/N0616.Demonstrate Computer Literacy	0	80	0	20	100	5
Total	500	360	90	100	1050	100

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Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.